

MH3100 Real Analysis I

Tutorial 5 Solutions

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Overview

- **Lecture coverage:** Lectures 13–17 — limit points, isolated points and closures (L13–14); closed sets (L15); extrema of closed and bounded sets and (sequential) compactness (L16); the Nested Set Property (L17).
- **Questions:** 8 questions (5 main + 3 Additional) from Tutorial 5; solutions extend the official Tutorial 5 “Answers and Hints.”
- **Notation:** $\mathbb{N} = \{1, 2, 3, \dots\}$ (0 is not a natural number in this course); closure of A is $\overline{A} = \overline{A}$. **This course never uses open sets, open covers, or the Heine–Borel Theorem:** compactness is always (*sequentially*) compact in the sense of the course’s own definition (every sequence in the set has a subsequence converging to a limit in the set).

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Question 1 (Limit points, isolated points, closures of five sets)

Problem

For each of the following sets, determine (i) the set of all its limit points; (ii) the set of all its isolated points; and (iii) its closure.

(a) $A = \left\{ (-1)^n + \frac{2}{n} : n \in \mathbb{N} \right\}.$

(b) $B = \{x \in \mathbb{Q} : 0 < x < 1\}.$

(c) $C = \{x \in \mathbb{R} : x \neq 0\}.$

(d) $D = \left\{ 1 + \frac{1}{4} + \frac{1}{9} + \cdots + \frac{1}{n^2} : n \in \mathbb{N} \right\}.$

(e) $E = \left\{ 1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n} : n \in \mathbb{N} \right\}.$

(Tutorial 5, Question 1; Lectures 13–15, limit points / isolated points / closure.)

Solution

Method 1: Sequential Criterion for Limit Points, part by part

Two reusable lemmas. (*Scratch work – not part of the written answer.*) Parts (a), (d) and (e) all involve the range of a strictly monotonic sequence. It pays to settle this once and for all, and likewise to settle how limit points behave under a union (needed for (a), whose set splits into an even-index and an odd-index piece).

Lemma 1 (ranges of strictly monotonic sequences). Let $\{c_k\}_{k \in \mathbb{N}}$ be a strictly monotonic sequence of real numbers, and let $C := \{c_k : k \in \mathbb{N}\}$.

- (i) Every $c_m \in C$ is an isolated point of C .
- (ii) If $c_k \rightarrow L \in \mathbb{R}$, then L is the unique limit point of C , and $L \notin C$.
- (iii) If $\{c_k\}$ is increasing and unbounded above (or decreasing and unbounded below), then C has no limit point at all.

Proof of Lemma 1. Without loss of generality take $\{c_k\}$ strictly decreasing (if it is strictly increasing, apply the same argument to the strictly decreasing sequence $\{-c_k\}$ and negate every conclusion).

(i). Fix $m \in \mathbb{N}$. Since $\{c_k\}$ is strictly decreasing, every index $k < m$ satisfies $c_k \geq c_{m-1} > c_m$, and every index $k > m$ satisfies $c_k \leq c_{m+1} < c_m$ (there is no lower constraint when $m = 1$). Let

$$\varepsilon_m := \begin{cases} \frac{1}{2}(c_1 - c_2) & \text{if } m = 1, \\ \frac{1}{2} \min\{c_{m-1} - c_m, c_m - c_{m+1}\} & \text{if } m \geq 2, \end{cases}$$

which is a positive number because $\{c_k\}$ is strictly decreasing. By the two displayed inequalities, every term c_k with $k \neq m$ satisfies $|c_k - c_m| \geq 2\varepsilon_m > \varepsilon_m$. Therefore $(c_m -$

$\varepsilon_m, c_m + \varepsilon_m) \cap C = \{c_m\}$, so c_m is not a limit point of C ; since $c_m \in C$, this means c_m is isolated in C .

(ii). Suppose $c_k \rightarrow L$. By the Monotone Convergence Theorem, $L = \inf\{c_k : k \in \mathbb{N}\}$, so $c_k \geq L$ for every k . If $c_{k_0} = L$ for some k_0 , then strict monotonicity gives $c_{k_0+1} < c_{k_0} = L$, contradicting L being a lower bound for $\{c_k\}$; hence $c_k > L$ strictly for every k , so $L \notin C$. The sequence $\{c_k\}$ itself therefore lies in $C \setminus \{L\}$ and converges to L , so by the Sequential Criterion for Limit Points, L is a limit point of C . No point other than L is a limit point of C . If $x = c_m \in C$ for some m , then by part (i) c_m is isolated in C , so x is not a limit point of C . Suppose instead $x \notin C$ and $x \neq L$; we show x is not a limit point of C . Let $\varepsilon_0 := |x - L|/2 > 0$. Since $c_k \rightarrow L$, there is $K \in \mathbb{N}$ such that $|c_k - L| < \varepsilon_0$ for every $k \geq K$, so every such c_k lies in the interval $(L - \varepsilon_0, L + \varepsilon_0)$, which does not contain x (as $|x - L| = 2\varepsilon_0$). Hence any c_k within ε_0 of x must satisfy $k < K$, i.e. lie among the finitely many terms $c_1, \dots, c_{K-1}$; since $x \notin C$, none of these finitely many terms equals x , so $\delta := \min(\{\varepsilon_0\} \cup \{|c_k - x| : k < K\}) > 0$, a minimum of finitely many positive numbers. Then $(x - \delta, x + \delta) \cap C = \emptyset$: no c_k with $k < K$ lies in it, by the choice of δ , and no c_k with $k \geq K$ lies in it either, since $\delta \leq \varepsilon_0$ and such $c_k \in (L - \varepsilon_0, L + \varepsilon_0)$, which is disjoint from $(x - \varepsilon_0, x + \varepsilon_0) \supseteq (x - \delta, x + \delta)$. Hence x is not a limit point of C . Combining both cases, L is the unique limit point of C .

(iii). Suppose $\{c_k\}$ is increasing and unbounded above (the decreasing/unbounded-below case is symmetric). Fix $x \in \mathbb{R}$. Since $\{c_k\}$ is unbounded above, there is $N \in \mathbb{N}$ with $c_N > x + 1$; since $\{c_k\}$ is increasing, $c_k > x + 1$ for every $k \geq N$, so the set $F := \{k \in \mathbb{N} : |c_k - x| \leq 1\} \subseteq \{1, \dots, N - 1\}$ is finite. If $F = \emptyset$, take $\varepsilon := 1$. Otherwise, since $\{c_k\}$ is strictly increasing it is injective, so at most one $k \in F$ has $c_k = x$; the (finite, possibly empty) set $\{|c_k - x| : k \in F, c_k \neq x\}$ consists of positive numbers, so it has a minimum $\delta > 0$ (or take $\varepsilon := 1$ if this set is empty), and we take $\varepsilon := \delta/2 \leq 1/2$. In either case, every c_k with $k \notin F$ satisfies $|c_k - x| > 1 > \varepsilon$, and every $c_k \in F$ with $c_k \neq x$ satisfies $|c_k - x| \geq \delta > \varepsilon$. Hence $(x - \varepsilon, x + \varepsilon) \cap (C \setminus \{x\}) = \emptyset$, so x is not a limit point of C . As $x \in \mathbb{R}$ was arbitrary, C has no limit point. $\square$

Lemma 2 (limit points of a union). For $A, B \subseteq \mathbb{R}$, a number x is a limit point of $A \cup B$ if and only if x is a limit point of A or a limit point of B .

Proof of Lemma 2. ($\Leftarrow$) If x is a limit point of A , there is a sequence $\{y_n\}$ in $A \setminus \{x\} \subseteq (A \cup B) \setminus \{x\}$ with $y_n \rightarrow x$; by the Sequential Criterion for Limit Points, x is a limit point of $A \cup B$. The case where x is a limit point of B is identical.

($\Rightarrow$) Suppose x is a limit point of $A \cup B$. By the Sequential Criterion for Limit Points, there is a sequence $\{y_n\}$ in $(A \cup B) \setminus \{x\}$ with $y_n \rightarrow x$. Every y_n lies in A or in B , so $\mathbb{N} = \{n : y_n \in A\} \cup \{n : y_n \in B\}$; since $\mathbb{N}$ is infinite and a union of two finite sets is finite, at least one of these two index sets is infinite. Say $\{n : y_n \in A\}$ is infinite (the other case is identical with B in place of A); listing its elements in increasing order gives a subsequence $\{y_{n_k}\}$ contained in $A \setminus \{x\}$. By the Theorem on Subsequential Limits, $y_{n_k} \rightarrow x$ as well, so by the Sequential Criterion for Limit Points, x is a limit point of A . $\square$

(a) $A = \{(-1)^n + 2/n : n \in \mathbb{N}\}$. (*Scratch work – not part of the written answer.*) Split A by the parity of n . For odd $n = 2k - 1$, the term is $-1 + 2/(2k - 1)$, strictly decreasing in k and $\rightarrow -1$. For even $n = 2k$, the term is $1 + 2/(2k) = 1 + 1/k$, strictly decreasing in k and $\rightarrow 1$. Note $n = 1$ gives the value 1 exactly, so $1 \in A$.

Proof. Let $c_k := -1 + \frac{2}{2k-1}$ and $d_k := 1 + \frac{2}{2k} = 1 + \frac{1}{k}$ for $k \in \mathbb{N}$, so that $A = A_{\text{odd}} \cup A_{\text{even}}$ where $A_{\text{odd}} := \{c_k : k \in \mathbb{N}\}$ and $A_{\text{even}} := \{d_k : k \in \mathbb{N}\}$. Both $\{c_k\}$ and $\{d_k\}$ are strictly decreasing (since $2/(2k-1)$ and $1/k$ are strictly decreasing in k), with $c_k \rightarrow -1$ and $d_k \rightarrow 1$. By Lemma 1(ii), $\text{LimPts}(A_{\text{odd}}) = \{-1\}$ with $-1 \notin A_{\text{odd}}$, and $\text{LimPts}(A_{\text{even}}) = \{1\}$ with $1 \notin A_{\text{even}}$. By Lemma 2,

$$\text{LimPts}(A) = \text{LimPts}(A_{\text{odd}}) \cup \text{LimPts}(A_{\text{even}}) = \{-1, 1\}.$$

Since $c_1 = 1$, we have $1 = c_1 \in A_{\text{odd}} \subseteq A$; and since $-1 \notin A_{\text{odd}}$ and $-1 \notin A_{\text{even}}$ (as $d_k > 1 > -1$ for all k), we have $-1 \notin A$. Hence the isolated points of A are

$$\text{IsolatedPts}(A) = A \setminus \text{LimPts}(A) = A \setminus \{-1, 1\} = A \setminus \{1\},$$

and the closure is

$$\overline{A} = A \cup \text{LimPts}(A) = A \cup \{-1, 1\} = A \cup \{-1\}.$$

□

(a) Limit points $\{-1, 1\}$; isolated points $A \setminus \{1\}$; $\overline{A} = A \cup \{-1\}$.

Beyond the hints. The official hint states the answer but not that 1 is *both* attained (at $n = 1$) *and* a limit point (as the limit of the even subsequence); a student who checks only “is 1 attained?” without also checking “is 1 approached by other points of A ?” will mis-classify it as isolated.

(b) $B = \mathbb{Q} \cap (0, 1)$. (*Scratch work – not part of the written answer.*) Expect $\text{LimPts}(B) = [0, 1]$ by density of $\mathbb{Q}$: every real number, not just those in $(0, 1)$, is approximated by rationals in $(0, 1)$ once we clip the approximating rationals into $(0, 1)$.

Proof. We first record a fact about $\mathbb{Q}$ that the hints assume but do not supply: for any reals $a < b$, there is a rational number in (a, b) . By the Archimedean Property, choose $n \in \mathbb{N}$ with $1/n < b - a$. By the Archimedean Property again, the set $\{k \in \mathbb{Z} : k > na\}$ is nonempty and bounded below (by na), so it has a least element k_0 (well-ordering, applied to the shifted set of natural numbers $\{k - \lceil na \rceil + 1 : k > na\}$). Minimality gives $k_0 - 1 \leq na$, so $k_0/n \leq a + 1/n < a + (b - a) = b$; and $k_0 > na$ gives $k_0/n > a$. Hence $q := k_0/n \in \mathbb{Q}$ satisfies $a < q < b$.

$\text{LimPts}(B) = [0, 1]$. Fix $x \in [0, 1]$. For each $n \in \mathbb{N}$, apply the displayed fact to the interval $(\max\{x - \frac{1}{n}, 0\}, \min\{x + \frac{1}{n}, 1\})$ (nonempty for every $n \in \mathbb{N}$: its two endpoints coincide only if $x - 1/n \geq 0$ and $x + 1/n \leq 1$ force $x = 0 = 1$ at once, which is impossible) to obtain a rational q_n in this interval; then $q_n \in (0, 1) \cap \mathbb{Q} = B$, $|q_n - x| < 1/n$, and $q_n \neq x$ can always be arranged by shrinking the interval slightly further whenever the constructed q_n happens to equal x (there remain infinitely many rationals strictly between $\max\{x - 1/n, 0\}$ and q_n , and the displayed fact supplies one). This produces a sequence $\{q_n\}$ in $B \setminus \{x\}$ with $q_n \rightarrow x$ (by the Squeeze-type bound $|q_n - x| < 1/n \rightarrow 0$), so by the Sequential Criterion for Limit Points, x is a limit point of B . Hence $[0, 1] \subseteq \text{LimPts}(B)$.

Conversely, if $x < 0$, let $\varepsilon := |x|/2 > 0$; every $b \in B \subseteq (0, 1)$ satisfies $b > 0 > x + \varepsilon$ (since $x + \varepsilon = x/2 < 0$), so $(x - \varepsilon, x + \varepsilon) \cap B = \emptyset$ and x is not a limit point of B . If $x > 1$, the symmetric argument with $\varepsilon := (x - 1)/2$ applies. Hence $\text{LimPts}(B) \subseteq [0, 1]$, and so $\text{LimPts}(B) = [0, 1]$.

Isolated points and closure. Since $B \subseteq (0, 1) \subseteq [0, 1]$, $\text{IsolatedPts}(B) = B \setminus \text{LimPts}(B) = B \setminus [0, 1] = \emptyset$, and $\overline{B} = B \cup [0, 1] = [0, 1]$. $\square$

(b) Limit points $[0, 1]$; isolated points $\emptyset$; $\overline{B} = [0, 1]$.

Beyond the hints. The hint gives no construction at all. The subtlety filled in above is that the approximating rationals must be *clipped* into $(0, 1)$ (so that they lie in B , not merely in $\mathbb{Q}$) even when x is an endpoint of $[0, 1]$.

(c) $C = \mathbb{R} \setminus \{0\}$.

Proof. Fix $x \in \mathbb{R}$. The equation $x + 1/n = 0$ has at most one solution $n \in \mathbb{N}$ (namely $n = -1/x$, when this happens to be a positive integer), so there is $N \in \mathbb{N}$ such that $x + 1/n \neq 0$ for every $n \geq N$. Then the sequence $\{x + 1/n\}_{n \geq N}$ lies in $C \setminus \{x\}$ (each term is nonzero, hence in C , and differs from x since $1/n \neq 0$) and converges to x (as $1/n \rightarrow 0$). By the Sequential Criterion for Limit Points, x is a limit point of C . Since $x \in \mathbb{R}$ was arbitrary, $\text{LimPts}(C) = \mathbb{R}$. As $C \subseteq \mathbb{R}$, $\text{IsolatedPts}(C) = C \setminus \mathbb{R} = \emptyset$, and $\overline{C} = C \cup \mathbb{R} = \mathbb{R}$. $\square$

(c) Limit points $\mathbb{R}$; isolated points $\emptyset$; $\overline{C} = \mathbb{R}$.

(d) $D = \{1 + \frac{1}{4} + \cdots + \frac{1}{n^2} : n \in \mathbb{N}\}$. (*Scratch work – not part of the written answer.*) D is the range of the partial-sum sequence of $\sum 1/n^2$. We must first know this series converges (the course does *not* give its value, so we only assert existence of the sum).

Proof. Let $s_n := \sum_{k=1}^n 1/k^2$, so $D = \{s_n : n \in \mathbb{N}\}$. For $n \geq 2$, $\frac{1}{n^2} < \frac{1}{n(n-1)} = \frac{1}{n-1} - \frac{1}{n}$, so

$$s_n = 1 + \sum_{k=2}^n \frac{1}{k^2} < 1 + \sum_{k=2}^n \left(\frac{1}{k-1} - \frac{1}{k} \right) = 1 + \left(1 - \frac{1}{n} \right) = 2 - \frac{1}{n} < 2$$

by telescoping. So $\{s_n\}$ is increasing (each term $1/n^2 > 0$) and bounded above by 2; by the Monotone Convergence Theorem, $\{s_n\}$ converges to some $s := \sum_{n=1}^{\infty} 1/n^2 = \sum_{n=1}^{\infty} 1/n^2 \in \mathbb{R}$. (The course has not established the value of s – in particular $s = \pi^2/6$ is not course material – so we only use that s exists as a real number.) Since $\{s_n\}$ is strictly increasing and converges to s , Lemma 1(ii) gives $\text{LimPts}(D) = \{s\}$ and $s \notin D$. Hence $\text{IsolatedPts}(D) = D \setminus \{s\} = D$ (as $s \notin D$), and $\overline{D} = D \cup \{s\}$. $\square$

(d) Limit points $\{s\}$, $s = \sum_{n=1}^{\infty} 1/n^2$; isolated points D ; $\overline{D} = D \cup \{s\}$.

(e) $E = \{1 + \frac{1}{2} + \cdots + \frac{1}{n} : n \in \mathbb{N}\}$. (*Scratch work – not part of the written answer.*) E is the range of the harmonic partial sums, which diverge to $+\infty$; Lemma 1(iii) then gives no limit points at all.

Proof. Let $t_n := \sum_{k=1}^n 1/k$, so $E = \{t_n : n \in \mathbb{N}\}$. We first show $\{t_n\}$ diverges, via the Cauchy Criterion for Series. Fix $N \in \mathbb{N}$ and take $n = N$, $m = 2N$. The tail sum has N terms, each at least $1/(2N)$:

$$\sum_{k=N+1}^{2N} \frac{1}{k} \geq N \cdot \frac{1}{2N} = \frac{1}{2} > \frac{1}{3}.$$

Since this holds for every $N \in \mathbb{N}$, no N makes the tail sums eventually smaller than $\varepsilon = 1/3$, so the Cauchy Criterion for Series fails for $\sum 1/n$; hence $\sum_{n=1}^{\infty} 1/n$ diverges, i.e. $\{t_n\}$ does not converge. Since $\{t_n\}$ is increasing (each term $1/n > 0$), the contrapositive of the Monotone Convergence Theorem shows $\{t_n\}$ is unbounded above (a bounded increasing sequence would converge). By Lemma 1(iii) (applied to the strictly increasing, unbounded-above sequence $\{t_n\}$), E has no limit point: $\text{LimPts}(E) = \emptyset$. Hence $\text{IsolatedPts}(E) = E \setminus \emptyset = E$, and $\overline{E} = E \cup \emptyset = E$. $\square$

(e) Limit points $\emptyset$; isolated points E ; $\overline{E} = E$.

Beyond the hints. Both (d) and (e) need the reader to notice that D and E are the ranges of *partial-sum sequences*, and to separately establish convergence (via the Comparison Test / telescoping, for $\sum 1/n^2$) or divergence (via the Cauchy Criterion for Series, for the harmonic series) before Lemma 1 can even be applied; the hint states only the conclusions.

Common slip. A limit point of a set need not belong to the set (as with -1 for A , or s for D); conversely a point of the set can still be a limit point of that same set, as 1 is for A . Students who assume “isolated” and “in the set” are the same notion mis-answer part (a).

Set	Limit points	Isolated points	Closure
A	$\{-1, 1\}$	$A \setminus \{1\}$	$A \cup \{-1\}$
B	$[0, 1]$	$\emptyset$	$[0, 1]$
C	$\mathbb{R}$	$\emptyset$	$\mathbb{R}$
D	$\{s\}$	D	$D \cup \{s\}$
E	$\emptyset$	E	E

Question 2 ((Sequential) compactness of five sets)

Problem

Determine if each of the following sets is (sequentially) compact.

- (a) $A = [0, 1] \cup [5, 6]$.
- (b) $B = \mathbb{R}_0^+$.
- (c) $C = \{x \in \mathbb{R} : 0 \leq x \leq 1 \text{ and } x \text{ is irrational}\}$.
- (d) $D = \left\{1 + \frac{1}{4} + \frac{1}{9} + \cdots + \frac{1}{n^2} : n \in \mathbb{N}\right\}$.
- (e) $E = \left\{1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n} : n \in \mathbb{N}\right\}$.

(Tutorial 5, Question 2; Lectures 16–17, Sequential Compactness / Extrema of Closed and Bounded Sets; builds on Question 1.)

Solution

Method 1: definition of compactness and the Sequential Compactness of Closed and Bounded Sets

- (a) $A = [0, 1] \cup [5, 6]$.

Proof. We show A is closed and bounded, then invoke the Sequential Compactness of Closed and Bounded Sets. Boundedness is immediate: $A \subseteq [0, 6]$. For closedness, let $\{a_n\}$ be a convergent sequence in A with limit a . Every term lies in $[0, 1]$ or in $[5, 6]$, so (as in Lemma 2's proof) at least one of these two index sets is infinite; say infinitely many $a_n \in [0, 1]$, giving a subsequence $\{a_{n_k}\} \subseteq [0, 1]$ with $a_{n_k} \rightarrow a$ (Theorem on Subsequential Limits). Since $0 \leq a_{n_k} \leq 1$ for every k , the Order Limit Theorem gives $0 \leq a \leq 1$, so $a \in [0, 1] \subseteq A$. The case where infinitely many $a_n \in [5, 6]$ is identical (giving $a \in [5, 6] \subseteq A$). Either way $a \in A$, so by the Sequential Criterion for Closed Sets, A is closed. Being closed and bounded, A is (sequentially) compact by the Sequential Compactness of Closed and Bounded Sets. $\square$

(a) A is compact.

- (b) $B = \mathbb{R}_0^+ = [0, \infty)$.

Proof. B is unbounded, so it cannot be compact: consider the sequence $b_n := n \in B$. Any subsequence $\{b_{n_k}\}$ satisfies $n_k \geq k$ (indices of a subsequence are strictly increasing), so by the Archimedean Property, for every $M > 0$ there is k with $n_k \geq k > M$; hence $\{b_{n_k}\}$ is unbounded and so diverges (a convergent sequence is bounded, by the Boundedness of Convergent Sequences). Thus $\{b_n\}$ has no convergent subsequence at all, so B fails the definition of (sequential) compactness. $\square$

(b) B is not compact.

- (c) $C = \{x \in [0, 1] : x \text{ irrational}\}$.

Proof. Let $x_n := \frac{\sqrt{2}}{n+2}$ for $n \in \mathbb{N}$. Each x_n is irrational: if $x_n = p/q$ for integers p, q with $q \neq 0$, then $\sqrt{2} = p(n+2)/q \in \mathbb{Q}$, contradicting the irrationality of $\sqrt{2}$. Also $0 < x_n \leq \sqrt{2}/3 < 1$, so $x_n \in C$ for every n . The sequence $\{x_n\}$ converges to 0 (as $n+2 \rightarrow \infty$), and 0 is rational, so $0 \notin C$. Since $\{x_n\}$ converges to 0, every subsequence of it also converges to 0 (Theorem on Subsequential Limits); as $0 \notin C$, no subsequence of $\{x_n\}$ converges to a point of C . Hence C fails the definition of (sequential) compactness. $\square$

(c) C is not compact.

(d) D (partial sums of $\sum 1/n^2$).

Proof. By Question 1(d), the sequence of partial sums $\{s_n\}$ lies in D and converges to $s = \sum_{n=1}^{\infty} 1/n^2 \notin D$. By the Theorem on Subsequential Limits, every subsequence of $\{s_n\}$ also converges to $s \notin D$, so $\{s_n\}$ is a sequence in D with no subsequence converging to a point of D . Hence D is not (sequentially) compact. $\square$

(d) D is not compact.

(e) E (partial sums of the harmonic series).

Proof. By Question 1(e), the sequence of partial sums $\{t_n\}$ is strictly increasing and unbounded above. As in part (b), any subsequence $\{t_{n_k}\}$ is unbounded above (given $M > 0$, unboundedness gives N with $t_N > M$, and then $n_k \geq N$ eventually forces $t_{n_k} \geq t_N > M$ by monotonicity), hence divergent by the Boundedness of Convergent Sequences. So $\{t_n\}$ has no convergent subsequence, and E is not (sequentially) compact. $\square$

(e) E is not compact.

Beyond the hints. The official hint gives only the five bare answers (“Compact”, “Not compact” $\times 4$) with no justification at all; every argument above – the closed-and-bounded argument for (a), the explicit unbounded-witness sequences for (b) and (e), and the explicit non-convergent-in-the-set witnesses reusing Question 1’s limit points for (c) and (d) – is supplied here in full.

Common slip. Exhibiting *one* sequence with no subsequence converging *in the set* is enough to rule out compactness; exhibiting a convergent sequence whose limit lies outside the set (as in (c), (d)) automatically forces every subsequence’s limit outside the set too, by the Theorem on Subsequential Limits – this is the fast route, and re-proving unboundedness from scratch (as would be needed for the “closed and bounded” criterion) is unnecessary once a witnessing sequence is in hand.

Set	(Sequentially) compact?
$A = [0, 1] \cup [5, 6]$	Yes
$B = \mathbb{R}_0^+$	No (unbounded)
C	No (not closed: irrationals $\rightarrow 0 \notin C$)
D	No (not closed: partial sums $\rightarrow s \notin D$)
E	No (unbounded)

Question 3 ($\sup A \in \overline{A}$ via the ε -Criterion for Closures)

Problem

Use the ε -Criterion for Closures to prove that if $A \subseteq \mathbb{R}$ is nonempty and bounded above, then $\sup A \in \overline{A}$. (*Tutorial 5, Question 3; Lecture 14, ε -Criterion for Closures; Lecture 1–3, ε -Criterion for Suprema.*)

Solution

Method 1: ε -Criterion for Suprema feeds directly into the ε -Criterion for Closures

Idea. (*Scratch work – not part of the written answer.*) The ε -Criterion for Closures asks, for every $\varepsilon > 0$, for a point of A within ε of $s := \sup A$. The ε -Criterion for Suprema already supplies a point of A strictly greater than $s - \varepsilon$; since it is also $\leq s$, it is automatically within ε of s .

Proof. Let $A \subseteq \mathbb{R}$ be nonempty and bounded above, and let $s := \sup A$ (which exists by the Axiom of Completeness). We verify the ε -Criterion for Closures for the number s : we must show that for every $\varepsilon > 0$, there exists $x \in A$ with $|x - s| < \varepsilon$.

Fix $\varepsilon > 0$. Since s is the *least* upper bound of A , the number $s - \varepsilon < s$ is not an upper bound for A (by the ε -Criterion for Suprema). Hence there exists $x \in A$ with $x > s - \varepsilon$. Since s is an upper bound for A , also $x \leq s$. Combining, $s - \varepsilon < x \leq s$, so $|x - s| = s - x < \varepsilon$. This produces, for every $\varepsilon > 0$, a point $x \in A$ with $|x - s| < \varepsilon$, which is exactly the ε -Criterion for Closures. Therefore $s \in \overline{A}$. $\square$

$\sup A \in \overline{A}$ for every nonempty, bounded-above $A \subseteq \mathbb{R}$.

Beyond the hints. The official hint states only “show the criterion for $s = \sup A$ implies the criterion for $s \in \overline{A}$ ” without ever writing either criterion down or checking that the point produced by the first ($x > s - \varepsilon$) satisfies the second ($|x - s| < \varepsilon$); the one-line reason ($x \leq s$ combined with $x > s - \varepsilon$) is supplied above.

Common slip. It is tempting to write $|x - s| < \varepsilon$ directly from $x > s - \varepsilon$ alone; this only gives $s - x < \varepsilon$ from below and needs the separate fact $x \leq s$ (so $s - x \geq 0$) to conclude $|x - s| = s - x < \varepsilon$ rather than merely $s - x > -\varepsilon$.

Question 4 (Closed sets under union and arbitrary intersection)

Problem

Use the Sequential Criterion for Closed Sets to prove the following statements.

- (a) If F_1 and F_2 are closed sets, then so is their union $F_1 \cup F_2$.
- (b) If $\{F_t\}_{t \in T}$ is an arbitrary collection of closed sets, then so is their intersection

$$\bigcap_{t \in T} F_t := \{a : \forall t \in T \ a \in F_t\}.$$

(Tutorial 5, Question 4; Lecture 15, Sequential Criterion for Closed Sets; (a) reuses the “shuffled sequence” technique of Tutorial 3, Question 3.)

Solution

Method 1: Sequential Criterion for Closed Sets

(a) $F_1 \cup F_2$ is closed. (*Scratch work – not part of the written answer.*) A convergent sequence in $F_1 \cup F_2$ has each term in F_1 or in F_2 ; by pigeonhole, infinitely many terms land in one of the two sets, giving a subsequence there with the same limit (this is exactly the argument used for a “shuffled” sequence $a_1, b_1, a_2, b_2, \dots$ in Tutorial 3, Question 3: any subsequence shares a further subsequence with one of the two constituent sequences).

Proof. Let $\{c_n\}$ be a convergent sequence contained in $F_1 \cup F_2$, with limit c . Every term satisfies $c_n \in F_1$ or $c_n \in F_2$, so $\mathbb{N} = \{n : c_n \in F_1\} \cup \{n : c_n \in F_2\}$; since $\mathbb{N}$ is infinite, at least one of these two index sets is infinite (a union of two finite sets is finite). Suppose $\{n : c_n \in F_1\}$ is infinite (the other case is identical with F_2 in place of F_1). Listing its elements in increasing order gives a subsequence $\{c_{n_k}\}$ contained entirely in F_1 . By the Theorem on Subsequential Limits, $\{c_{n_k}\}$ converges to the same limit c . Since F_1 is closed and $\{c_{n_k}\}$ is a convergent sequence in F_1 , the Sequential Criterion for Closed Sets gives $c \in F_1 \subseteq F_1 \cup F_2$. In the other case, the identical argument gives $c \in F_2 \subseteq F_1 \cup F_2$. Either way, $c \in F_1 \cup F_2$. Since $\{c_n\}$ was an arbitrary convergent sequence in $F_1 \cup F_2$, the Sequential Criterion for Closed Sets shows $F_1 \cup F_2$ is closed. $\square$

$$F_1, F_2 \text{ closed} \implies F_1 \cup F_2 \text{ closed.}$$

(b) $\bigcap_{t \in T} F_t$ is closed.

Proof. Let $\{a_n\}$ be a convergent sequence contained in $\bigcap_{t \in T} F_t$, with limit a . Fix an arbitrary $t \in T$. Since $\bigcap_{t \in T} F_t \subseteq F_t$, the sequence $\{a_n\}$ is also a convergent sequence contained in F_t . Since F_t is closed, the Sequential Criterion for Closed Sets gives $a \in F_t$. As $t \in T$ was arbitrary, $a \in F_t$ for every $t \in T$, which is precisely the condition for $a \in \bigcap_{t \in T} F_t$. Since $\{a_n\}$ was an arbitrary convergent sequence in $\bigcap_{t \in T} F_t$, the Sequential Criterion for Closed Sets shows $\bigcap_{t \in T} F_t$ is closed. $\square$

$$\{F_t\}_{t \in T} \text{ all closed} \implies \bigcap_{t \in T} F_t \text{ closed.}$$

Beyond the hints. The official hint for (a) only gestures at the analogy with the shuffled-sequence question and never explains how “sharing a subsequence with F_1 or F_2 ” concludes the *limit* lies in $F_1 \cup F_2$; the missing step is exactly the appeal to closedness of F_1 (or F_2) via the Sequential Criterion, supplied above. For (b), the hint states the one-line observation ($a \in \bigcap F_t \Rightarrow a \in F_t$ for every t) but never runs it in the direction needed (starting from a convergent sequence in the intersection and concluding the limit is in each F_t , hence in the intersection); this is written out above.

Common slip. In (b), T may be infinite (even uncountable); the proof never needs to extract a subsequence because the same fixed sequence $\{a_n\}$ already lies in every single F_t simultaneously – there is no pigeonhole step here, unlike in (a).

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Question 5 (A closed subset of a compact set is compact)

Problem

Use the definition of (sequential) compactness to prove that a closed subset of a (sequentially) compact set is (sequentially) compact; i.e., for any compact set $K \subseteq \mathbb{R}$ and any subset $A \subseteq K$, prove that A is compact if it is closed. **Do not use the Sequential Compactness of Closed and Bounded Sets or the Bolzano-Weierstrass Theorem.**

[Note: A proof without using the Sequential Compactness of Closed and Bounded Sets or the Bolzano-Weierstrass Theorem is applicable to compact sets in $\mathbb{Q}$ (or any ordered field) for which these properties do not hold.] (*Tutorial 5, Question 5; Lecture 16, definition of (sequential) compactness; Lecture 15, Sequential Criterion for Closed Sets.*)

Solution

Method 1: definition of compactness directly

Idea. (*Scratch work – not part of the written answer.*) A sequence in A is automatically a sequence in K (as $A \subseteq K$), so K 's compactness gives a subsequence converging in K ; closedness of A then pulls that limit back into A , using that the subsequence itself never left A .

Proof. Let $K \subseteq \mathbb{R}$ be (sequentially) compact and let $A \subseteq K$ be closed. Let $\{a_n\}$ be an arbitrary sequence contained in A . Since $A \subseteq K$, $\{a_n\}$ is also a sequence contained in K . Since K is (sequentially) compact, by the definition of (sequential) compactness, $\{a_n\}$ has a subsequence $\{a_{n_k}\}$ converging to some limit $x \in K$.

Now $\{a_{n_k}\}$ is a subsequence of $\{a_n\}$, hence is itself a sequence contained in A (every term a_{n_k} is a term of the original sequence $\{a_n\} \subseteq A$). So $\{a_{n_k}\}$ is a convergent sequence contained in A , with limit x . Since A is closed, the Sequential Criterion for Closed Sets gives $x \in A$.

We have shown: the arbitrary sequence $\{a_n\}$ contained in A has a subsequence $\{a_{n_k}\}$ converging to a limit $x \in A$. This is precisely the definition of (sequential) compactness for A . Hence A is (sequentially) compact. $\square$

$$\boxed{K \text{ compact and } A \subseteq K \text{ closed} \implies A \text{ compact.}}$$

Beyond the hints. The official hint says to “compare the definitions” without carrying out the chain of reasoning; the two steps that must be made explicit are (1) a sequence in A is, without further work, a sequence in K (since $A \subseteq K$), which is what lets K 's compactness apply at all; and (2) the subsequence produced by K 's compactness is *still* a sequence inside A (not just inside K), which is what lets A 's closedness pull the limit back into A rather than leaving it in $K \setminus A$.

Common slip. This proof never uses that $K \subseteq \mathbb{R}$ specifically (no boundedness, no Bolzano–Weierstrass, no Axiom of Completeness beyond what is baked into K 's compactness) – exactly as flagged in the question, this is why the same argument transfers

verbatim to compact subsets of $\mathbb{Q}$ or any ordered field, where the Sequential Compactness of Closed and Bounded Sets can fail.

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Question A1 (Closed sets via distance-zero points)

Problem

For a nonempty set $A \subseteq \mathbb{R}$ and a number $a \in \mathbb{R}$, we define the *distance of a from A* as the infimum of the set $\{|x - a| : x \in A\}$, which is nonempty and bounded below. Prove that a nonempty set $A \subseteq \mathbb{R}$ is closed if and only if every number that has distance 0 from A is in A . (*Tutorial 5, Additional Question 1; Lecture 14, ε -Criterion for Closures; Lecture 1–3, ε -Criterion for Infima; builds on Question 3.*)

Solution

Method 1: ε -Criterion for Infima $\leftrightarrow$ ε -Criterion for Closures

Idea. (*Scratch work – not part of the written answer.*) “Distance 0” is an infimum statement about $\{|x - a| : x \in A\}$; unwind it with the ε -Criterion for Suprema and Infima (infimum case; dual to the supremum case used in Question 3), and notice the resulting condition on $x \in A$ is literally the ε -Criterion for Closures for $a \in \bar{A}$.

Proof. Let $A \subseteq \mathbb{R}$ be nonempty, and for $a \in \mathbb{R}$ write $d(a, A) := \inf\{|x - a| : x \in A\}$ for the distance of a from A (which exists by the Axiom of Completeness, as the set $\{|x - a| : x \in A\}$ is nonempty and bounded below by 0).

Claim: $d(a, A) = 0$ if and only if $a \in \bar{A}$. Suppose $d(a, A) = 0$. Fix $\varepsilon > 0$. By the ε -Criterion for Suprema and Infima (infimum case: since 0 is the *greatest* lower bound of $\{|x - a| : x \in A\}$, the number $0 + \varepsilon$ is not a lower bound), there exists $x \in A$ with $|x - a| < 0 + \varepsilon = \varepsilon$. This holds for every $\varepsilon > 0$, which is exactly the ε -Criterion for Closures, so $a \in \bar{A}$. Conversely, suppose $a \in \bar{A}$. Fix $\varepsilon > 0$; by the ε -Criterion for Closures, there exists $x \in A$ with $|x - a| < \varepsilon$. Since also $|x - a| \geq 0 =$ (the trivial lower bound of $\{|x - a| : x \in A\}$), this produces, for every $\varepsilon > 0$, an element of $\{|x - a| : x \in A\}$ that is $< 0 + \varepsilon$; together with 0 being a lower bound, the ε -Criterion for Suprema and Infima (infimum case) gives $\inf\{|x - a| : x \in A\} = 0$, i.e. $d(a, A) = 0$. This proves the claim.

($\Rightarrow$) A closed $\implies$ every distance-0 point is in A . Suppose A is closed and $d(a, A) = 0$. By the claim, $a \in \bar{A}$. Since A is closed, by the Closure Criterion for Closed Sets $\bar{A} \subseteq A$, so $a \in A$.

($\Leftarrow$) every distance-0 point in $A \implies A$ closed. Suppose every $a \in \mathbb{R}$ with $d(a, A) = 0$ lies in A . Let $a \in \bar{A}$ be arbitrary; by the claim, $d(a, A) = 0$, so by hypothesis $a \in A$. Hence $\bar{A} \subseteq A$; since also $A \subseteq \bar{A}$ always, $\bar{A} = A$, so by the Closure Criterion for Closed Sets, A is closed. $\square$

$$A \text{ closed} \iff (d(a, A) = 0 \implies a \in A) \text{ for every } a \in \mathbb{R}.$$

Beyond the hints. The hint’s step (a) refers to “the approach for the preceding question” without restating the ε -Criterion for Suprema and Infima (infimum case; the dual of the supremum case used in Question 3, not previously written out for infima in this tutorial); step (b) asserts the two closure conditions “are the same condition on x ” without showing the algebra ($|x - a| < \varepsilon \iff$ the $\{|x - a|\}$ -set has an element $< 0 + \varepsilon$); and step (c) never assembles the final iff from (a)+(b), all supplied above.

Common slip. $d(a, A) = 0$ does *not* mean some $x \in A$ satisfies $|x - a| = 0$ (that would already say $a \in A$, trivialising the problem); it only means such x can be found *arbitrarily close*, which is exactly why the ε -Criterion (not a single exact witness) is the right tool.

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Question A2 (Disjoint closed sets can have distance 0)

Problem

For two sets $A, B \subseteq \mathbb{R}$, we define the *distance* between them as the infimum of the set $\{|a - b| : a \in A, b \in B\}$. If two closed sets A and B are disjoint (i.e., $A \cap B = \emptyset$), must the distance between them necessarily be positive? (*Tutorial 5, Additional Question 2; Lecture 13–15, closed sets and distance to a set; contrasts with Additional Question 3, the compact case.*)

Solution

Method 1: an explicit counterexample

Idea. (*Scratch work – not part of the written answer.*) Want two closed sets that never touch but sneak arbitrarily close. Take the integers $A = \mathbb{N}$ and a second sequence of non-integers B that creeps toward A without ever landing on it: $b_n = n + 1/n$ is within $1/n$ of the integer n , yet is never itself an integer.

Proof. Lemma (uniformly separated sets are closed). If $S \subseteq \mathbb{R}$ is such that there exists $\delta > 0$ with $|s - s'| \geq \delta$ for all distinct $s, s' \in S$, then S has no limit point (hence is vacuously closed).

Proof of Lemma. Suppose, for contradiction, that $x \in \mathbb{R}$ is a limit point of S . By the Sequential Criterion for Limit Points, there is a sequence $\{a_k\}$ in $S \setminus \{x\}$ with $a_k \rightarrow x$. Applying the definition of convergence with $\varepsilon = \delta/4 > 0$, there is $K \in \mathbb{N}$ such that $|a_k - x| < \delta/4$ for all $k \geq K$. Then for $k, l \geq K$, $|a_k - a_l| \leq |a_k - x| + |x - a_l| < \delta/2 < \delta$; since distinct elements of S are at least δ apart, this forces $a_k = a_l$ for all $k, l \geq K$, i.e. the sequence is eventually equal to some fixed value $c \in S$. But then $x = \lim a_k = c \in S$, contradicting $a_k \neq x$ for every k (which forces $c \neq x$). Hence no such x exists, and S has no limit point; a set with no limit points contains all of them vacuously, so S is closed. $\square$

The counterexample. Let $A := \mathbb{N} = \{1, 2, 3, \dots\}$ and $B := \{n + 1/n : n \in \mathbb{N}, n \geq 2\}$. $A \cap B = \emptyset$: every element of A is an integer, while every element of B is $n + 1/n$ with $n \geq 2$, which is strictly between the consecutive integers n and $n + 1$ (since $0 < 1/n \leq 1/2 < 1$), hence not an integer.

A is closed: distinct integers differ by at least 1, so the Lemma applies with $\delta = 1$.

B is closed: for $n \geq 2$, the map $n \mapsto n + 1/n$ is strictly increasing, since

$$(n+1) + \frac{1}{n+1} - \left(n + \frac{1}{n}\right) = 1 - \left(\frac{1}{n} - \frac{1}{n+1}\right) = 1 - \frac{1}{n(n+1)} \geq 1 - \frac{1}{6} = \frac{5}{6} > 0$$

using $n(n+1) \geq 6$ for $n \geq 2$. So distinct elements $b_n, b_{n'} \in B$ (say $n < n'$) satisfy $b_{n'} - b_n \geq b_{n+1} - b_n \geq 5/6$ (each step of the strictly increasing sequence contributes at least $5/6$). Hence the Lemma applies to B with $\delta = 5/6$.

Distance is 0: for each $n \geq 2$, taking $a = n \in A$ and $b = n + 1/n \in B$ gives $|a - b| = 1/n$, so

$$0 \leq \inf\{|a - b| : a \in A, b \in B\} \leq \frac{1}{n} \quad \text{for every } n \geq 2.$$

Fix $\varepsilon > 0$; by the Archimedean Property choose $n \geq 2$ with $1/n < \varepsilon$. Then the pair $(a, b) = (n, n + 1/n)$ shows $\{|a - b| : a \in A, b \in B\}$ has an element $< \varepsilon$; combined

with 0 being a lower bound, the ε -Criterion for Suprema and Infima (infimum case) gives $\inf\{|a - b| : a \in A, b \in B\} = 0$.

Thus A and B are disjoint closed sets with distance 0. $\square$

No: $A = \mathbb{N}$ and $B = \{n + 1/n : n \geq 2\}$ are disjoint, closed, and at distance 0.

Beyond the hints. The official hint gives only the answer “No”; the counterexample, and the verification that both sets are closed (via the uniform-gap Lemma, not previously needed elsewhere in this tutorial), are entirely supplied above.

Common slip. It is not enough to exhibit two disjoint sets with distance 0: both sets must actually be *closed*, which needs its own proof (here, via the gap Lemma) – an open interval endpoint pair such as $(0, 1)$ and $\{0\}$ would also have distance 0, but $(0, 1)$ is not closed, so it would not answer the question as posed.

Question A3 (Disjoint compact sets have positive, attained distance)

Problem

Recall that for two sets $A, B \subseteq \mathbb{R}$, we define the *distance* between them as the infimum of the set $\{|a - b| : a \in A, b \in B\}$.

Prove that if two (sequentially) compact sets K_1 and K_2 are disjoint, then the distance between them is positive and is attained by two numbers $a_1 \in K_1$ and $a_2 \in K_2$; i.e., if the sets K_1 and K_2 are (sequentially) compact and have an empty intersection, then there are two numbers $a_1 \in K_1$ and $a_2 \in K_2$ for which the distance between K_1 and K_2 is $|a_1 - a_2|$, which is positive. (*Tutorial 5, Additional Question 3; Lecture 16, Sequential Compactness and Extrema of Closed and Bounded Sets; Lecture 4–6, triangle inequality in the limit sense; contrasts with Additional Question 2.*)

Solution

Method 1: compactness of the distance set, then Extrema of Closed and Bounded Sets

Idea. (*Scratch work – not part of the written answer.*) Standing assumption: K_1, K_2 are nonempty (otherwise the two numbers a_1, a_2 the problem asks for could not exist). Package the whole set of pairwise distances $S = \{|a - b| : a \in K_1, b \in K_2\}$ as a single subset of $\mathbb{R}$, show it is compact by chasing subsequences through K_1 then K_2 , and let the Extrema of Closed and Bounded Sets supply its minimum.

Proof. Assume $K_1, K_2 \subseteq \mathbb{R}$ are nonempty, (sequentially) compact, and disjoint. Let

$$S := \{|a - b| : a \in K_1, b \in K_2\}.$$

Step 1: a limit lemma. If $\{p_m\}$ and $\{q_m\}$ are sequences of reals with $p_m \rightarrow p$ and $q_m \rightarrow q$, then $|p_m - q_m| \rightarrow |p - q|$. *Proof:* Fix $\varepsilon > 0$. Since $p_m \rightarrow p$, there is M_1 with $|p_m - p| < \varepsilon/2$ for $m \geq M_1$; since $q_m \rightarrow q$, there is M_2 with $|q_m - q| < \varepsilon/2$ for $m \geq M_2$. For $m \geq \max\{M_1, M_2\}$, the (reverse) triangle inequality gives

$$\begin{aligned} \left| |p_m - q_m| - |p - q| \right| &\leq |(p_m - q_m) - (p - q)| \\ &= |(p_m - p) - (q_m - q)| \\ &\leq |p_m - p| + |q_m - q| < \varepsilon. \end{aligned}$$

By the definition of limits of sequences, $|p_m - q_m| \rightarrow |p - q|$. □

Step 2: S is (sequentially) compact. Let $\{x_n\}$ be an arbitrary sequence in S . For each n , choose (by definition of S) $a_n \in K_1$ and $b_n \in K_2$ with $x_n = |a_n - b_n|$. Since K_1 is compact, $\{a_n\}$ has a subsequence $\{a_{n_k}\}$ converging to some $a \in K_1$. Now consider $\{b_{n_k}\}$, a sequence in K_2 ; since K_2 is compact, it has a further subsequence $\{b_{n_{k_j}}\}$ converging to some $b \in K_2$. The corresponding subsequence $\{a_{n_{k_j}}\}$ of $\{a_{n_k}\}$ still converges to a (Theorem on Subsequential Limits, applied to the already-convergent sequence $\{a_{n_k}\}$). By Step 1,

$$x_{n_{k_j}} = |a_{n_{k_j}} - b_{n_{k_j}}| \longrightarrow |a - b| \quad \text{as } j \rightarrow \infty.$$

Since $a \in K_1$ and $b \in K_2$, the limit $|a - b|$ lies in S by definition of S . So $\{x_{n_{k_j}}\}$ is a subsequence of $\{x_n\}$ converging to a point of S . As $\{x_n\}$ was an arbitrary sequence in S , this shows S is (sequentially) compact by the definition.

Step 3: $0 \notin S$, so $S \subseteq (0, \infty)$. If $0 \in S$, there would exist $a \in K_1, b \in K_2$ with $|a - b| = 0$, i.e. $a = b$; then $a \in K_1 \cap K_2$, contradicting disjointness. So $0 \notin S$. Since every element of S is of the form $|a - b| \geq 0$, and 0 itself is excluded, every element of S is strictly positive.

Step 4: conclude via Extrema of Closed and Bounded Sets. By Step 2, S is (sequentially) compact; by the Sequential Compactness of Closed and Bounded Sets, S is therefore closed and bounded. Since S is also nonempty (as K_1, K_2 are nonempty, any $a \in K_1, b \in K_2$ give $|a - b| \in S$), the Extrema of Closed and Bounded Sets gives that S has a minimum, $m := \min S$. Since $m \in S$, by Step 3, $m > 0$. By definition of S , there exist $a_1 \in K_1$ and $a_2 \in K_2$ with $m = |a_1 - a_2|$. Since $m = \min S = \inf S$, and $\inf S$ is exactly the distance between K_1 and K_2 as defined in the problem, we conclude that the distance between K_1 and K_2 equals $|a_1 - a_2| = m > 0$, attained at $a_1 \in K_1, a_2 \in K_2$. $\square$

$$\boxed{\text{dist}(K_1, K_2) = |a_1 - a_2| = \min S > 0 \text{ for some } a_1 \in K_1, a_2 \in K_2.}$$

Beyond the hints. The official hint leaves out the actual subsequence chase in Step 2 (extracting through K_1 , then through K_2 , then re-using the Theorem on Subsequential Limits to keep the K_1 -side subsequence's limit fixed); it also never states or proves the limit lemma of Step 1 in full ε - N form, and never spells out why $0 \notin S$ forces the minimum to be strictly positive (Step 3), which is the crux of “positive” in the conclusion.

Common slip. Two separate applications of compactness are needed here (once for K_1 , once for K_2), and the second extraction must be applied to the subsequence *already* produced by the first, not to a fresh copy of $\{a_n\}$; otherwise the two resulting subsequences of a - and b -terms would not share the same index set $\{n_{k_j}\}$, and Step 1's limit lemma could not be applied to the paired terms $|a_{n_{k_j}} - b_{n_{k_j}}|$.

Method Map

Q.	Core topic	Methods included
1	Limit points / isolated points / closure of five sets	Sequential Criterion for Limit Points; two reusable lemmas (isolated ranges of monotonic sequences; limit points of a union); convergence of $\sum 1/n^2$ (Comparison Test / telescoping) vs. divergence of $\sum 1/n$ (Cauchy Criterion for Series); density of $\mathbb{Q}$ via the Archimedean Property.
2	(Sequentially) compact or not	Sequential Criterion for Closed Sets + Sequential Compactness of Closed and Bounded Sets (part (a)); explicit witnessing sequences reusing Question 1 (parts (c)–(e)); unboundedness witness (part (b)).
3	$\sup A \in \overline{A}$	ε -Criterion for Suprema feeding the ε -Criterion for Closures.
4	Closed sets under union / arbitrary intersection	Sequential Criterion for Closed Sets; pigeonhole + shuffled-sequence technique (union); direct membership chase (arbitrary intersection).
5	Closed subset of a compact set is compact	Definition of (sequential) compactness directly, plus Sequential Criterion for Closed Sets; no boundedness or Bolzano–Weierstrass used.
A1	Closed sets via distance-zero points	ε -Criterion for Infima $\iff \varepsilon$ -Criterion for Closures; Closure Criterion for Closed Sets.
A2	Disjoint closed sets, distance 0	Explicit counterexample ($\mathbb{N}$ and $\{n + 1/n\}$); uniform-gap lemma for closedness.
A3	Disjoint compact sets, positive attained distance	(Sequential) compactness of the distance set $\{ a - b \}$ (double subsequence extraction); triangle-inequality limit lemma; Extrema of Closed and Bounded Sets.

Technique Summary

- Strictly monotonic sequences have isolated, sparse ranges.** A strictly monotonic sequence's range consists entirely of isolated points, with at most one extra limit point (its limit, if it converges) or none at all (if it diverges to $\pm\infty$). This single lemma settles Question 1(a),(d),(e) and Additional Question 2's counterexample.
- Limit points of a union are the union of limit points.** Proved once via pigeonhole on the index set and the Theorem on Subsequential Limits; reused for Question 1(a) and again (in the closed-sets guise) for Question 4(a).
- ε -Criteria for suprema/infima translate directly into ε -Criteria for closures.** A point produced “just inside” an extremum automatically lies in an ε -neighbourhood of that extremum – this is the engine behind Question 3 and Additional Question 1.
- Compactness chains through operations that preserve sequences.** Distances built from two compact sets (Additional Question 3) inherit compactness by extracting one subsequence per factor, in sequence, not in parallel.

5. **To disprove compactness, one witnessing sequence suffices.** A single sequence with no subsequence converging in the set (because the whole sequence's limit, shared by every subsequence, lies outside the set, or because the sequence is unbounded) is enough, and is usually faster than checking closedness and boundedness separately.

Final Answers

1. See the summary table at the end of Question 1: A – limit points $\{-1, 1\}$, isolated $A \setminus \{1\}$, closure $A \cup \{-1\}$. B – limit points, isolated set, and closure $[0, 1]$, $\emptyset$, $[0, 1]$. C – $\mathbb{R}$, $\emptyset$, $\mathbb{R}$. D – $\{s\}$ (where $s = \sum_{n=1}^{\infty} 1/n^2$), D , $D \cup \{s\}$. E – $\emptyset$, E , E .
 2. (a) Compact. (b) Not compact. (c) Not compact. (d) Not compact. (e) Not compact.
 3. $\sup A \in \overline{A}$ for every nonempty, bounded-above $A \subseteq \mathbb{R}$.
 4. (a) F_1, F_2 closed $\implies F_1 \cup F_2$ closed. (b) An arbitrary intersection of closed sets is closed.
 5. A closed subset of a (sequentially) compact set is (sequentially) compact.
- A1.** A is closed if and only if every number at distance 0 from A lies in A .
- A2.** No: $A = \mathbb{N}$ and $B = \{n + 1/n : n \in \mathbb{N}, n \geq 2\}$ are disjoint, closed, and at distance 0.
- A3.** The distance between disjoint compact sets K_1 and K_2 is positive and attained at some $a_1 \in K_1$, $a_2 \in K_2$: writing $m := \text{dist}(K_1, K_2)$, we have $m = |a_1 - a_2| > 0$.